Language features under weight quantization

Activation agreement and intervention effects
27 languages, three quantizers and two Gemma sizes

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Completed evidence

- Selected language-feature groups have **measurable next-token effects** beyond activity-matched controls in all 27 languages of both Gemma sizes.
- **Better quantization improves preservation and transfer**. GPTQ reduces the changes seen under the original RTN recipe.
- Activation agreement, language association and intervention effects provide **different measurements**.
- The paper reports positive effects alongside a comprehension null and limited repair. Most analyses are exploratory.

Source: Course report revision 4; reports/FINAL_RESEARCH_QUESTIONS.md

Final synthesis questions

1. How do selectivity and resource tier relate to activation preservation across quantizers?
2. What do language-group interventions change across tasks and precision?
3. How informative are activation survival and association retention about intervention effects?
4. How do context, downstream weights and generic sensitivity affect intervention transfer?

These organize the completed work retrospectively. Original frozen hypotheses and verdicts remain part of the record.

Source: reports/FINAL_RESEARCH_QUESTIONS.md

Models, text and interventions

- Primary: Gemma-2-2B, layer 12, Gemma Scope SAE with 16,384 features.
- Extensions: Gemma-2-9B with its layer-20 Gemma Scope SAE; Llama-3.2-1B with a TopK SAE.
- FLORES: 27 languages, 300 aligned evaluation sentences each. Belebele: 18 languages, 385 questions each. Fresh Wikipedia transfer panel: six languages.
- Remove or scale the top 64 selected features; compare with activity-matched non-selective groups. Preserve the SAE reconstruction error.
- Model inference bfloat16, SAE and edits float32. Sham and weight-restore identities check the edit path.

Source: Functional, transfer and 9B freezes; course report Sections 3–4

Three quantization recipes

Legacy RTN	Original rounding implementation, with attention scales shared across heads.
Matched RTN	Per-output-channel layout, using the same grid as GPTQ. No calibration.
GPTQ	Calibration-based error compensation, with 8,192 Wikipedia tokens per language.

Same 2B model, dictionary, texts, edits and arithmetic. Rounded weights stay in floating-point storage. INT16 is a small real perturbation reference.

Source: reports/gptq_extension/RESULTS.md; Frantar et al., GPTQ (2022)

RQ1: Selectivity and resource tier

- Original RTN selective-minus-control survival: -1.44 **percentage points at INT6**, Holm-adjusted $p = 0.128$. Non-rejection does not establish equivalence.
- INT6 survival for low/mid/high tiers: **69.4/90.6/95.1%**. Conditioning on activation scale attenuates the initial tier association.
- Matched RTN changes the significance label of small contrasts; GPTQ shows no selective penalty in those tests.
- Preservation depends on the quantizer and feature activity. Resource tiers are observational categories.

Source: reports/phase1_summary.md; reports/gptq_extension/results/stage1/

GPTQ improves activation and behavioral preservation

INT4, Gemma-2-2B	Legacy RTN	Matched RTN	GPTQ
All-active survival	12.7%	18.6%	**30.5%**
NLL increase, nats/token	0.590	0.385	**0.130**

GPTQ improves both metrics in **27 of 27 languages**. Its mean survival gain over legacy RTN is **17.83 points [16.29,19.38]** at INT4.

Intervals resample paired languages. Calibration and compensation change together.

Source: reports/gptq_extension/results/stage1/stage1_comparison.md

What the survival correlation measures

An exact least-squares identity separates gain, noise and signal scale:

$$b = ga + c + e, \qquad \lambda = \log(R_1^{-2} - 1) = \log \sigma_e^2 - 2 \log |g| - 2 \log \sigma_a.$$

- Streamed statistics cover **214,506 active language–feature cells**; correlation reproduction within 3×10^{-8}.
- Across six cells, scale correlation is about -0.51 **for** λ, versus -0.004 **to** $+0.114$ for threshold-normalized encoder perturbation ν.
- This clarifies activation preservation. It does not imply that correlation or SAEs provide no information.

Source: reports/survival_measurement/stage3/stage3_results.md

RQ2: Selected groups matter for predicting text

- Gemma-2-2B FP excess ablation harm is positive with a lower interval above zero in **27 of 27 languages**.
- Median excess harm: **0.262 nats/token**. Mean: 0.704 [0.684,0.723], with sentence-bootstrap uncertainty.
- At INT6 the effect stays near FP. At INT4 the median per-language effect ratio is **1.54 under legacy RTN**, 1.36 under matched RTN, **1.22 under GPTQ**.
- These are causal effects of defined edits. A larger removal effect measures changed dependence on the group.

Source: Functional results/eval/report.md; GPTQ stage3 analysis_gptq_eval/report.md

Graded suppression and limited gain repair

- Deeper suppression increases net harm in **all 27 languages**. FP median effects: 0.022, 0.090, 0.169, 0.262 nats/token.
- Dose-response and consistency rules hold; the strict legacy precision-ordering rule fails because of an intermediate-width dip.
- Main fine-gain repair criteria fail. A fresh-sentence RTN follow-up finds a small $1.1\times$ benefit, about **0.00135 nats/token** (20/27 languages).
- The main 300-sentence boosting hint weakens under GPTQ. State-defined groups are not more consequential, and shrinkage does not predict the optimal gain.

Source: reports/functional_validation/gain_study/; reports/gptq_extension/RESULTS.md

Gemma-2-9B supports the group precision pattern

Original RTN	Gemma-2-2B	Gemma-2-9B
FP positive excess harm	27/27	27/27
INT6 median effect ratio	1.031	1.019
INT4 median effect ratio	1.538	**1.586**
INT4 95% language interval	[1.281,2.629]	[1.238,2.264]

Paired model-difference intervals include zero. The review reproduces the raw-file analysis byte-identically. Two flawed frozen criteria are documented separately. The 9B GPTQ result is not part of this evidence.

Source: reports/functional_validation/gemma9b/REVIEWED_SUMMARY.md

The Llama check shows heterogeneity

- Llama-3.2-1B FP median excess harm is **0.048 [0.018,0.173]** nats/token, positive in 21/27 languages.
- Its frozen consistency rule requires 24/27, so the recorded verdict is **not consistent**.
- The INT4 median effect ratio is **0.86 [0.74,1.03]**; the Gemma amplification pattern does not carry over.
- Model family, dictionary, depth, size and tokenizer change together. The discrepancy bounds generalization and preserves the measured Gemma contrasts.

Source: reports/functional_validation/llama/report.md; llama_robust/

Comprehension and patch recovery measure different outcomes

- Belebele FP excess correct-option loss: -0.0001 **nats/item [**$-0.0035, +0.0031$**]**. The required FP effect is not established.
- Accuracy is a separate endpoint: about 42.6% FP. GPTQ INT4 reaches 31.8%, above 25% chance, with a small group-ablation change.
- Non-selective control sets reduce accuracy by 1.2–2.2 points; other-language selective controls reach 6.0 points.
- INT4 donor recovery: selected K64 group **2–3% of the loss gap**, full residual **35–41%**. These patch recovery shares do not partition causal responsibility.

Source: Functional mc_eval/report.md, rescue/report.md; GPTQ stage3/

RQ3: Functional identity can persist despite activation drift

Legacy RTN INT4, F5 sample of 1,487 features	Measurement
Pass activation-correlation threshold	13.2%
Median effect-fingerprint cosine	**0.839 [0.829,0.848]**
Own-feature identity retrieval	**99.7%**
Meet INT16-derived fidelity convention	0.3%

Features retain recognizable effects while their exact effects change. GPTQ raises median INT4 effect cosine to **0.923**.

Source: reports/functional_validation/results/f5/report.md; GPTQ extension F5

Activation survival adds modest positive predictive information

- On 1,412 reliable features, adding continuous activation correlation improves language-held-out effect prediction by **3.1–4.4%** across all six Gemma cells.
- Every 95% language-bootstrap interval is above zero. Legacy INT4: **3.7% [2.3,5.2]**; GPTQ INT4: **3.5% [2.1,5.0]**.
- These positive gains fall below the project's 10% practical convention. Gain and ν provide no resolved additional improvement.
- Llama INT4 repeats a modest contribution: **4.2% [1.6,6.7]**. Binary association labels are less informative.

Source: reports/survival_effect_rescue/stage4_gemma_report_2026-10-28.md; llama/

RQ4: Crossed context and downstream weights

For the same FP-amplitude edit, compare four model states:

	FP suffix	Quantized suffix
FP prefix	D_{FF}	D_{FQ}
Quantized prefix	D_{QF}	D_{QQ}

$$C = D_{QF} - D_{FF}, \quad W = D_{FQ} - D_{FF}, \quad I = D_{QQ} - D_{QF} - D_{FQ} + D_{FF}.$$

The vector identity is exact. Term norms and medians need not add. A fifth cell uses the quantized model's own amplitude.

Source: reports/intervention_transfer/results/main/; course report Section 8

Intervention transfer improves with the quantizer

INT4 eligible feature/context items	Legacy	Matched	GPTQ
Effect cosine	0.768	0.822	**0.877**
Normalized error	0.664	0.572	**0.476**

- Legacy fixed group harm: 0.848 versus 0.263 FP. The native-amplitude edit gives 0.433.
- Group loss emphasizes the context term; individual effect vectors emphasize downstream weights. The endpoint and edit definition matter.
- Control directions share substantial drift. Effect-size matching finds a modest extra component on a small selected subset.

Source: GPTQ results/stage2/; transfer results/main/ and recovery/

Prediction and fresh-text transfer

- Generic sensitivity reduces main-panel INT4 prediction error by **21.5% [12.7,29.4]**. INT6: 10.9% [-1.2,21.4], unresolved.
- A similar predictor works at INT16. Its value concerns generic intervention sensitivity, beyond quantization alone.
- Fresh Wikipedia INT4 effect cosine: **0.754/0.808/0.876** for legacy/matched/GPTQ. The quantizer ordering repeats.
- External prediction criteria hold at both widths for RTN; GPTQ meets INT4 but not INT6. Averaging fingerprints raises apparent transfer agreement.

Source: Transfer predictive_validation.csv; GPTQ RESULTS.md, external panel

Interpretation and shared methodological limits

- SAE features depend on dictionary training, sparsity and size. Incomplete recovery and mixed features remain open field-wide issues.
- Defined interventions yield causal effects in the tested model. A complete natural circuit account requires additional evidence.
- Most analyses reuse previously examined sentences. Frozen stage choices improve transparency without creating independent confirmation.
- Fixed model/layer/dictionary settings bound external validity. Quantizer comparisons and the 9B/Llama checks already provide useful evidence about that boundary.

Source: Gao et al. (2024); Gemma Scope (2024); course report limitations

Answers to the four synthesis questions

1. **Preservation depends on quantizer and scale**. A universal selective-destruction account is not established.

2. **Groups contribute to text prediction**. Behavioral relevance varies by task and edit; main gain repair remains limited.

3. **Activation and effect preservation share modest information**. Recognizable functional identity survives substantial activation drift.

4. **Portability is measurable and quantizer-dependent**. Context/weight contrasts and generic controls clarify where transfers change.

Source: reports/FINAL_RESEARCH_QUESTIONS.md; completed-study manuscript

Evidence and completion boundary

- The completed study includes the GPTQ extension, measurement decomposition and reviewed 9B original-RTN evidence.
- Gemma-2-9B GPTQ contributes no result yet. Additional quantizer or precision audits are extensions of the completed work.
- Sources, populations, endpoints and uncertainty are mapped in the report's `evidence_map.md`.

Code: `https://github.com/TH4KUR/always-late`

Artifacts: `https://huggingface.co/datasets/karma-skz/always-late-artifacts`

Run summaries: `https://wandb.ai/karma-skz-/anlp-project`